F1+TF1 Computer Networks

Case Study 1

1. Form a group of up to three students for the lab-based project. The same group of students can work on the case study.
2. Select base paper title from the link given in Excel sheet and it must be unique (two batches are not allowed to take same title)
3. https://docs.google.com/spreadsheets/d/1sHZ4u_-HXnuCgQAYjFd5v47_hVJdIdqKUbVpy4qLH2s/edit?usp=sharing
4. The title of the case study should be the same as the title of your base paper. Include the names and registration numbers of all your project teammates in the document.
5. Do Literature Survey for 20 papers similar to selected base paper title. It must be in both table format and paragraph format.

   Table Format

S.No	Title of the paper, Author name, year	Methodology	Key Findings	Advantage and Limitations

   Paragraph format sample

   Yesodha et al (2024) introduce an elliptic curve encryption-based energy-efficient secured Ant Colony Optimization (ACO) routing protocol for WSNs. The protocol combines the advantages of elliptic curve encryption for data security with the energy-efficient capabilities of ACO to optimize routing paths. It ensures secure data transmission while minimizing energy consumption, addressing the challenges of energy constraints and security in WSNs. This approach enhances network lifetime, improves security, and ensures reliable communication in dynamic and resource-limited environments like WSNs.

   Feroz Khan (2024) presented an Enhanced Multi-Attribute Based Trusted Attack Resistance (EMBTR) model for secure routing of sensor nodes in WSNs. The model incorporates multiple attributes, such as energy levels, trustworthiness, and node behavior, to assess and select the most reliable routing paths. EMBTR enhances network security by identifying and resisting attacks while ensuring efficient communication. The approach is designed to handle the unique challenges of WSNs, such as resource constraints and vulnerability to malicious nodes, offering a robust solution for secure data transmission in these networks.

Reference: (include reference for all 20 papers)

1. Yesodha, K., Krishnamurthy, M., Thangaramya, K., & Kannan, A. Elliptic curve encryption-based energy-efficient secured ACO routing protocol for wireless sensor networks. *Journal of Supercomputing, pp.* 1–34, 13(1), 2024.
2. Feroz Khan, A. B. An enhanced multi attribute based trusted attack resistance (EMBTR) for the secure routing of sensor nodes in wireless sensor network. *Wireless Personal Communication* pp.2397–2407, **137**(4), 2024.

pp. 2397–2407 is page number

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